

DERICK AARON LIEVIAN

Indonesian Citizen

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Education

The Chinese University of Hong Kong, Shenzhen

Sep. 2025 – Jun. 2029

Bachelor of Science in Computer Science and Engineering

Shenzhen, China

CGPA: 3.692

Technical Skills

Languages: Python, Java, HTML/CSS, JavaScript, TypeScript, SQL, NoSQL

Technologies/Frameworks: Spring Boot, Node.js, Express, NestJS, Next, React, MongoDB, Tailwind CSS, JavaFX, PostgreSQL, Redis, Git, GitHub, Linux, Docker

Projects

AI Learning Toolkit | *NextJS, Gemini API, LangChain, MongoDB, Supabase, Pinecone, Tailwind CSS* **March 2026**

- Built a full-stack study platform using NextJS and JavaScript that lets users upload PDFs to generate flashcards and ask questions about their documents.
- Created an automated pipeline using the Gemini API to break down long PDFs and extract structured flashcard decks instantly.
- Integrated LangChain and Pinecone vector databases to build a document QnA feature that pulls exact context from PDFs for accurate answers.
- Managed a hybrid backend setup, using Supabase for PDF file storage and user authentication, and MongoDB for user data and saved flashcards.

Multi-Tenant API Proxy | *Java, TypeScript, Spring Boot, React, PostgreSQL, Redis, Docker, Bucket4j* **August 2026**

- Designed and built a full-stack API gateway handling authentication, rate limiting, and request proxying for concurrent clients.
- Implemented JWT authentication with refresh tokens stored in HttpOnly cookies, securing user sessions against XSS attacks.
- Engineered a distributed rate limiter with Redis and Bucket4j, ensuring consistent rate limits across multiple service instances, and designed a threshold-based caching strategy with Redis.
- Containerized the entire application stack with Docker, simplifying local development and deployment.
- Built a responsive React dashboard with Tailwind CSS, enabling developers to manage APIs and view their proxy key.
- Created a long-lived Proxy key system with hashed storage for secure developer access.
- Developed a dynamic proxy engine with URL rewriting and header forwarding for multi-tenant API routing.

Claude Code Clone | *Python, OpenAI API, OpenRouter API* **August 2026**

- Built an AI coding assistant using Python and OpenRouter API that executes file operations and shell commands through natural language.
- Implemented multi-tool agent architecture (Read, Write, Bash) enabling autonomous file system interaction and code modifications.
- Engineered tool-calling loop system allowing the AI to handle complex multi-step tasks without human intervention.
- Integrated secure API key management with python-dotenv following industry security best practices.

Awards / Achievements

Half-Tuition Entry Scholarship

September 2025 – Present

Languages

English (Fluent), Mandarin (Limited working proficiency), Bahasa Indonesia (Native)